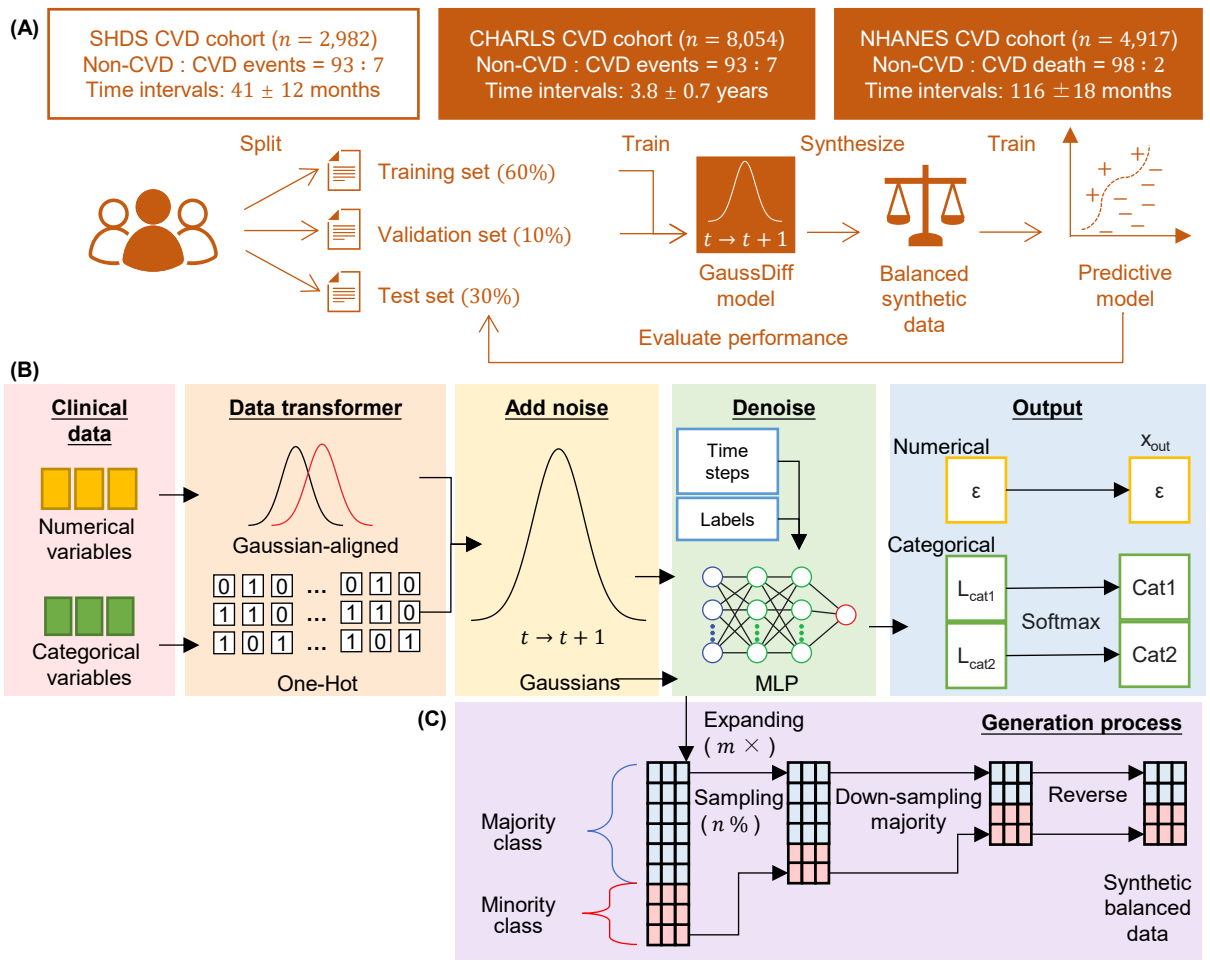

(A)
SHDS CVD cohort (n = 2,982)
Non-CVD : CVD events = 93 : 7
Time intervals: 41 ± 12 months
CHARLS CVD cohort (n = 8,054)
Non-CVD : CVD events = 93 : 7
Time intervals: 3.8 ± 0.7 years
NHANES CVD cohort (n = 4,917)
Non-CVD : CVD death = 98 : 2
Time intervals: 116 ±18 months
Split
Training set (60%)
Validation set (10%)
Test set (30%)
Train
$t \rightarrow t+1$
GaussDiff model
Synthesize
Balanced synthetic data
Train
Predictive model
Evaluate performance
(B)
Clinical data
Numerical variables
Categorical variables
Data transformer
Gaussian-aligned
One-Hot
Add noise
$t \rightarrow t+1$
Gaussians
Denoise
Time steps
Labels
MLP
Output
Numerical
x_{out}
ε
ε
Categorical
L_{cat1}
L_{cat2}
Softmax
Cat1
Cat2
(C)
Expanding (m ×)
Sampling (n %)
Majority class
Minority class
Down-sampling majority
Reverse
Generation process
Synthetic balanced data